



DATAVERSE X NUBRA • SESSION 1 OF 3

QUANT ENGINEERING 101

How machines took over the markets, and how you can build with them

ONCE UPON A TIME, IN 11TH STANDARD



Advanced Portfolio Management
With Nubra API
Friday, 31st July
11:30 AM - 12:30 PM
REGISTER NOW

Build a Delta-Neutral Hedging Bot That Rebalances Itself
With Nubra API
Sunday, 2nd Aug
12:00 PM - 01:00 PM
REGISTER NOW

Building a Gamma-Hedging Bot with Nubra API
Saturday, 12th Sep
12:00 PM - 1:00 PM

Build an Automated Smart Money Concepts Scanner using Nubra APIs
LIVE SESSION
FRIDAY, 10TH JUN

TODAY

"Wait, algo trading is a thing?"

Now: teaching it to rooms like this one.

WHAT HAPPENS TODAY

The day

SESSION 1

See it

- ◆ The story of markets: shouting halls to machines
- ◆ How a machine sees the market: the data types
- ◆ Nubra OSS, live
- ◆ Quiz: Dalal Street Showdown

SESSION 2

Build it

- ◆ UAT vs PROD: sandbox and live
- ◆ Live options-strategy demo, built with ChatGPT
- ◆ You build a quant screener
- ◆ Dashboards
- ◆ Quiz

SESSION 3

Test it

- ◆ Backtesting vocabulary
- ◆ Why backtests lie
- ◆ Pro engine demo
- ◆ 25-minute strategy competition, judged on hidden data

See it, build it, test it. Everything runs on a sandbox: no real money all day.

BEFORE WE GO FURTHER

Please read this. **Once.**

- ◆ This workshop is for **education only**. It is not investment advice or a recommendation to trade anything.
- ◆ Trading and investing involve **risk of loss**, including loss of all capital.
- ◆ Examples, charts and past results are **illustrative**. Past performance does not predict the future.
- ◆ Everything we build in these sessions runs on a **sandbox with no real money**.
- ◆ Views are the presenters' own, not those of BMSIT or any other organisation.

The **platform** behind today

- ◆ A trading platform for **stocks, futures and options**, built by **Zanskar Securities Pvt Ltd**
- ◆ A **SEBI-registered** broker with NSE, BSE, MCX and MSEI memberships
- ◆ Option chain, multi-leg strategy builder, live scanners, charts
- ◆ **OMS**, order management, built in-house: turns every request into one clean internal order (defaults, canonical fields, tags, basket and strategy metadata)
- ◆ **RMS**, risk management, built in-house: should this order be allowed **now**? What if it partially fills? Does it break a broker or exchange control?
- ◆ Around the exchange, Nubra's job is **stability**: reliable place, modify and cancel; safe reconnects, retries and session recovery



One of many doors

Code meets markets here. Builders, data engineers, researchers, traders. You do not have to trade to work in this field.

Sources: nubra.io (re-check the registration line on the day); "Trading APIs: what developers think happens vs what exchanges do" and "Inside the Exchange Black Box" (Nubra blog, by the presenter).

OPEN TO DEVELOPERS

Built for **builders**

REST API

Any language, plain HTTP

Python SDK

The fast path. Our path.

WebSockets

Live streaming data

UAT

Sandbox. Safe. No real money. **This is where we work.**

PROD

Live market. Real consequences.

You will install it yourself in Block 3. One line.

WHAT ACTUALLY HAPPENS WHEN YOU PRESS BUY

The **journey** of an order



Inside the cage: no humans, no screens. Only deterministic code, at microsecond scale. Better price wins, then earlier time wins.

Source: "Inside the Exchange Black Box" (Nubra blog, Feb 2026, by the presenter). Diagram: ours.

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THE MAP FOR THE NEXT TWO SESSIONS

Six things you can do

01 • Login

OTP login

02 • Reference

Instruments, indices

03 • Market data

Prices, quotes, option chain, history

04 • Real-time

Live quotes, Greeks, order book

05 • Trading

Place, modify, cancel, margin checks

06 • Portfolio

Positions, holdings, funds

BEFORE THE STORY STARTS

WHO'S IN THE ROOM?

backboard-crafter-closure.ngrok-free.dev

Register once: name · USN · a 4-digit PIN. Then 5 quick questions.

No right answers. Be honest. Your answers appear live.

BLOCK 2 · THE STORY

In NSE's cash market, what share of trading value is done by **algorithms**?

55%

55% of NSE cash-market turnover, FY26 (NSE Market Pulse)

More than half. A record, in FY26.

Source: NSE Market Pulse, April 2026 (algorithmic share of equity-cash turnover, FY26: 55%). "Algo" is NSE's tag for orders sent through algo terminals; it is not the same as high-frequency trading.

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HALL

>

SCREEN

>

MACHINE

FIRST, THE BASICS

A market is a **sabzi mandi**



A **share** is a tiny slice of a company. The stock market is a mandi where slices are traded.

ERA 1 · THE SHOUTING HALL

Shouting halls, everywhere

- ◆ **Open outcry:** shout your price, signal with your hands. No shout, no trade
- ◆ Chicago's pits, New York's floor, Bombay's ring: the same game
- ◆ **BSE:** founded 9 July 1875, Asia's first exchange; a trading ring until 1995



Photos: "Men working the floor at the Chicago Board of Trade", Stanley Kubrick for Look magazine, 1949, Library of Congress, public domain (Wikimedia Commons); "Commodities futures trading", Lars Plougmann, 7 May 1992, CC BY-SA 2.0 (Wikimedia Commons). BSE facts: Wikipedia, Equentis; re-check on bseindia.com.

THE HALL'S MOST FAMOUS STORY

Scam 1992: what really happened

- ◆ A Bombay broker, **Harshad Mehta**, used unbacked bank receipts to pull money out of banks into shares
- ◆ Size: roughly ₹4,000 to 5,000 crore
- ◆ Exposed by journalist **Sucheta Dalal** in the Times of India, **23 April 1992**
- ◆ Arrested Nov 1992. Market crashed. Reforms and transparent screen trading **accelerated**



The 2020 SonyLIV series, starring Pratik Gandhi, is based on the book by Sucheta Dalal and Debashis Basu.

Sources: Wikipedia (1992 Indian stock market scam; Scam 1992), iPleaders. Figures vary by source, hence a range. Poster: SonyLIV via Wikipedia (fair use); photo: Wikimedia Commons.

HALL

SCREEN

MACHINE

ERA 2 • THE SCREEN

Screens replaced shouting

NSE WENT LIVE ON SCREENS

3 Nov 1994

Equity trading on NEAT, screen-based from day one

BSE'S BOLT REPLACED THE RING

Mar 1995

BSE On-Line Trading; the shouting stopped



Same market. **New eyes.** Orders now travel as data, not as shouts. Once an order is data, a machine can send it.

Sources: NSE; Wikipedia (NSE); Equentis (BSE BOLT, 14 Mar 1995, secondary). BSE date: re-check on bseindia.com.

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ERA 2, PART TWO • THE RETAIL BOOM

Then it went **into your pocket**

- ◆ **2010s:** flat-fee discount brokers make trading cheap (₹0 delivery, a flat fee per order)
- ◆ Mobile apps and UPI make opening and funding an account easy

DEMAT ACCOUNTS

4.09 cr →
22+ cr

Mar 2020 → May 2026. About **5.6 times** in six years.
Accounts, not unique people.

Sources: CDSL/NSDL via NSE release (2020); Outlook Money (2026, NSDL 4.51 cr + CDSL 18.38 cr). Latest total is approximate: "over 22 crore". Discount-broker pricing: the brokers' own pricing pages (model from 2010).

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HALL >

SCREEN >

MACHINE

THE MANUAL ERA'S GREATEST HITS

People who **did it by hand**

Rakesh Jhunjhunwala

Started in **1985** with **₹5,000**.

Net worth at death (Aug 2022): **\$5.8 billion**, about ₹56,100 crore at today's rate. Mainly a long-term investor.



Radhakishan Damani

Began as a stock trader in the Harshad Mehta era, then built **DMart**.

Forbes, 30 Sep 2026: **\$20.4 billion**, about ₹1.97 lakh crore. Wealth is mostly from DMart.

RD

Radhakishan
Damani
b. 1955 ·
Mumbai

Decades of research, patience and nerve. **One human, one screen.**

Sources: Wikipedia, SCMP obituary, The News Minute (Jhunjhunwala); Forbes profile (Damani). Rupee figures at ₹96.76/US\$ (7 Oct 2026), approximate. Net worth is not income. Jhunjhunwala photo: Wikimedia Commons. No freely licensed photo of Damani exists on Commons, hence the name card.

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HALL

SCREEN

MACHINE

MEANWHILE, TODAY'S RETAIL TRADERS

87.7%

of Indian individual F&O traders **lost money** in FY26

Average loss

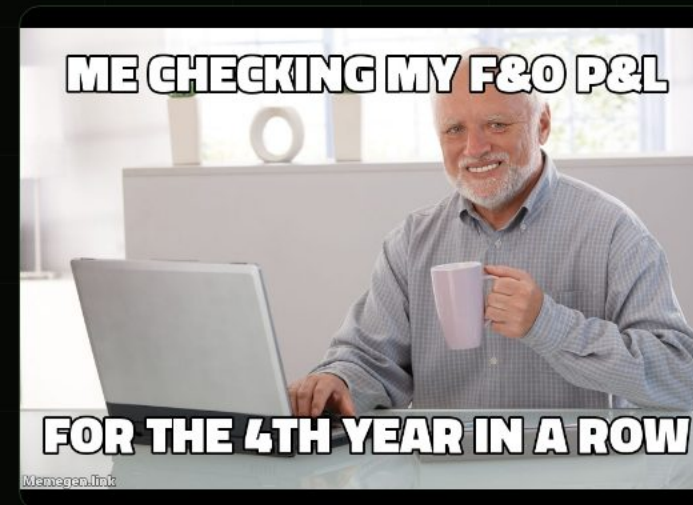
≈ ₹1.17 lakh per trader

Every year

about **9 in 10** lose, FY22 to FY26

Under 30s

89% lost money



Source: SEBI study, August 2026 (individual traders in equity F&O, FY26). Earlier SEBI studies: 93% (FY22-FY24), about 91% (FY25).

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SCREEN

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SAME SEBI STUDY. DIFFERENT GROUP.

99%

of proprietary-trader and foreign-investor gross profit came from firms
that **used algos**

Proprietary traders made about ₹44,000 crore in gross trading profit in FY26.

Source: SEBI study, August 2026. "Algo entities" = entities that placed at least one algorithmic order in the year. Gross profit, before costs.

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ERA 3 · THE MACHINE

They were mathematicians and coders

Edward Thorp

Wrote *Beat the Dealer* (1962): maths that beat casino blackjack. Then he took it to markets: 19.1% a year before fees, 1969 to 1988.

ET

Edward O.
Thorp
b. 1932 ·
mathematician

D. E. Shaw

Founded in 1988 by a **computer scientist** who left Columbia's CS faculty for Wall Street.

DS

David E. Shaw
b. 1951 ·
computer
scientist

Jim Simons

A mathematician. Founded Renaissance Technologies, home of the Medallion Fund. Died May 2024.



Not shouting. **Models, data and code.**

Sources: Wikipedia (Thorp, Shaw, Simons), deshaw.com, ValueWalk (Thorp returns, secondary). Simons photo: Wikimedia Commons. No freely licensed photos of Thorp or Shaw on Commons, hence name cards.

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HALL

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MACHINE

THE POINT OF THE WHOLE STORY

Best **manual** vs best **machine**

MANUAL • WARREN BUFFETT, BERKSHIRE
HATHAWAY

≈20%

a year, compounded, 1965 to 2023 (exactly 19.8%;
the S&P 500 did 10.2%)

MACHINE • JIM SIMONS, MEDALLION FUND

≈66%

a year **before fees**, 1988 to 2018 (about 39% after
fees)

MANUAL • RAKESH JHUNJHUNWALA,
LIFETIME

\$5.8B

net worth at death, August 2022 (Forbes)

MACHINE • MEDALLION, LIFETIME

≈\$104.5B

trading profits, 1988 to 2018 (reported)

Medallion's figures are **reported** (Zuckerman's book), not audited; Berkshire's come from its own annual letter. A net worth and a fund's profits are different things. The point is the gap, not the decimals.

Sources: Berkshire Hathaway 2023 shareholder letter, performance table (19.8% vs 10.2%, 1965-2023); Wealth Management and arXiv 2405.10917 for Medallion (from "The Man Who Solved the Market"); Wikipedia and The News Minute for Jhunjhunwala (\$5.8B, Forbes).

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WHY MACHINES CHANGED EVERYTHING

The manual era's **limits**, then four **superpowers**

One human, one screen

- ◆ **Slow:** about 250 ms to react
- ◆ **Emotional:** fear, FOMO, "one more trade"
- ◆ **Limited coverage:** a handful of charts
- ◆ **No testing:** you find out with real money

Speed

Machines work in microseconds: thousands of decisions in one blink.

Discipline

No panic, no FOMO. The rules run the same way every time.

Scale

Watch thousands of instruments at once.

Testable

Replay your rules on past data **before** risking a rupee. (Session 3.)

Rule: no trades after 3:15

Me: Ek Aur trade



Latency: human visual reaction about 250 ms (PubNub); HFT systems in microseconds (The RINSE, vendor claim). Order-of-magnitude comparison only.

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THE OTHER SIDE

Automation without risk control

- ◆ **1 Aug 2012:** Knight Capital deployed faulty software
- ◆ More than **4 million** unintended orders in about **45 minutes**
- ◆ Loss of roughly **\$440 to 460 million** ($\approx$ ₹4,300 to 4,450 crore)



A machine is a **faster way to be right**. And a faster way to be **wrong**.

Sources: SEC press release 2013-222; Knight Capital 10-Q (Aug 2012). Figures differ between the firm's filing (\$440M) and the SEC (over \$460M). Rupee figures at ₹96.76/US\$. Illustration: AI-generated.

IF YOU BUILD IN INDIA

There are **rules** for retail algos

Through a broker

You use the broker's API. The broker is accountable to the exchange. No open APIs.

Secure access

Registered static IP, OAuth login and two-factor authentication.

Higher speeds

Algos above an orders-per-second threshold must be registered with the exchange (reported as 10 per second).

Timeline

SEBI circulars of 4 Feb 2025 and 30 Sep 2025.
Applies to all brokers from **1 April 2026**.

Tonight: sandbox only. No real money. Read your broker's API terms before going live.

Sources: SEBI circulars (4 Feb 2025; 30 Sep 2025), read directly. The 10 orders-per-second figure is from broker explainers of NSE's circular, not SEBI's text.

HALL

>

SCREEN

>

MACHINE

FROM THE HALL TO THE MACHINE

THE HALL

shouting

THE SCREEN

orders as data

THE MACHINE

code decides

**So how does a machine
see the market?**

How does a machine **see** the market?

- ◆ Who is even **tradable**?
- ◆ What is the price **right now**?
- ◆ Who wants to **buy and sell**?
- ◆ What **happened before**?
- ◆ What are the **options**?
- ◆ What is happening **live**?



Download the **SDK**

```
# one line. this is the moment.  
python -m pip install nubra-sdk
```

Check it worked

```
python -c "import  
nubra_python_sdk"
```

No error = installed.

What you just got

Login, instruments, market data,
streaming, orders, portfolio: one
package.

Open source

MIT licence · 105 runnable
examples ·
[github.com/NubraAPI/nubra-
python-sdk](https://github.com/NubraAPI/nubra-python-sdk)

Get a **session** first

- ◆ Phone number + **OTP** + **MPIN**
- ◆ Done **once**; one client reused everywhere
- ◆ **UAT** = sandbox, **PROD** = live
- ◆ **Never put secrets on a slide or on GitHub**

```
from nubra_python_sdk.start_sdk import \
    InitNubraSdk, NubraEnv

nubra = InitNubraSdk(
    NubraEnv.UAT, env_creds=True)

# phone + MPIN live in a local .env file:
# PHONE_NO="000000000000"
# MPIN="0000"
```

Analogy: show your college ID plus a one-time gate code at the lab once. After that, you walk between rooms.

Every instrument has a **ref_id**

Exchange

The marketplace: NSE, BSE, MCX

Symbol

A short name: **HDFCBANK**

ref_id

Nubra's own ID for exactly **one** instrument

Like a **roll list**: names can repeat across batches, your enrolment number never does.

Look it up, **never guess**

```
from nubra_python_sdk.refdata.instruments import InstrumentData

instruments = InstrumentData(nubra)
hdfc = instruments.get_instrument_by_symbol(
    "HDFCBANK", exchange="NSE")
print(hdfc.ref_id)
```

You get

ref_id · stock_name · exchange · lot_size · tick_size
· expiry · strike_price · option_type

Gotcha

Resolve the ref_id from the symbol. Do not
hardcode it.

BOX 3 · THE PRICE RIGHT NOW

A **photo**, not a video

```
market_data = MarketData(nubra)

p = market_data.current_price(
    "NIFTY", exchange="NSE")
print(p.price, p.prev_close, p.change)
```

GOTCHA

100 paise = ₹1

Prices come back as integers in paise. Divide by 100 before showing anyone.

Fields: price · prev_close · change (% already). Snapshot only: it does not update on its own.



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BOX 4 • WHO WANTS TO BUY AND SELL?

The order book

ASK	₹ 1,004	400
ASK	₹ 1,003	250
ASK	₹ 1,002	900
spread • last traded ₹1,001.5		
BID	₹ 1,001	700
BID	₹ 1,000	1,200
BID	₹ 999	300

Illustrative numbers to explain the idea, not real market data.

- ◆ **Asks:** sellers, lowest price first
- ◆ **Bids:** buyers, highest price first
- ◆ Price moves when the two queues **meet**

Like a vegetable stall board: last sold at X, three want to pay Y, five will sell at Z.

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One call, five levels

```
ref_id = instruments.get_instrument_by_symbol(  
    "HDFCBANK", exchange="NSE").ref_id  
  
quote = market_data.quote(ref_id=ref_id, levels=5)  
print(quote.orderBook.bid[0], quote.orderBook.ask[0])
```

Needs

a `ref_id`, not a symbol

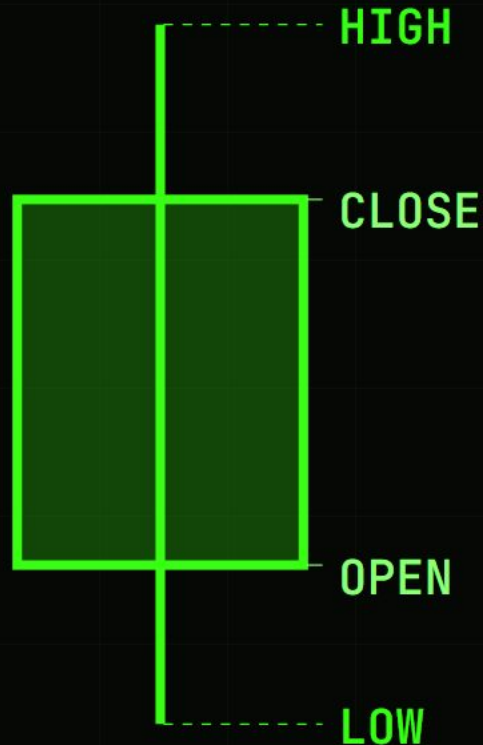
Returns

last price, last qty, volume, bid[],
ask[] (price, quantity,
num_orders)

Order

bids high→low, asks low→high

One candle = five numbers



- ◆ **Open:** first price of the period
- ◆ **High:** the highest price
- ◆ **Low:** the lowest price
- ◆ **Close:** last price of the period
- ◆ **Volume:** how many shares changed hands

Green = closed higher than it opened. Like one over of a cricket match: the first ball, best and worst moments, the final score, total runs.

Ask for any **window**

```
result = market_data.historical_data({  
    "exchange": "NSE", "type": "STOCK",  
    "values": ["ASIANPAINT"],  
    "fields": ["open", "high", "low", "close", "cumulative_volume"],  
    "startDate": "2025-04-19T11:01:57.000Z",  
    "endDate": "2025-04-24T06:13:57.000Z",  
    "interval": "1d", "intraDay": False, "realTime": False  
})
```

Dates

UTC

Prices

paise (÷100)

Volume

cumulative: use diff()

Limit

60 calls / min, 5
symbols each

BOX 5 · SAME MARKET, DIFFERENT ZOOM

Pick your timeframe

1s

1m

2m

3m

5m

15m

30m

1h

1d

1w

1mth

Short candles

Fast, noisy, lots of data. Minute data reaches back about 3 months.

Long candles

Calm, slow, goes back much further. Good for the big picture.

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EXCHANGE

Options **move** with the price

NIFTY • THE UNDERLYING

22,310

strike for both options: 22,250

22250 CE • CALL • RIGHT TO BUY
AT 22,250

₹183

intrinsic ₹60 + time value ₹123

22250 PE • PUT • RIGHT TO SELL
AT 22,250

₹123

intrinsic ₹0 + time value ₹123

UNDERLYING

STRIKE 22,250

CALL PRICE

PUT PRICE

Drifts on its own. Push it: drag the slider, or press ↑ / ↓. Model: intrinsic value + a time value that shrinks as the price moves away from the strike. Illustrative only.

A real **NIFTY** option chain

CALL OI	CALL LTP	STRIKE	PUT LTP	PUT OI
6,63,130	₹222.05	22,100	₹89.85	38,81,280
4,42,195	₹190.70	22,150	₹107.45	20,64,075
32,60,725	₹160.05	22,200	₹128.65	67,84,570
32,78,015	₹135.35	22,250 · ATM	₹151.50	22,93,915
66,71,470	₹111.85	22,300	₹177.45	54,98,870
31,91,305	₹90.50	22,350	₹206.05	11,92,555
1,07,13,755	₹72.00	22,400	₹240.10	31,64,850

Spot 22,231.80

ATM = the strike nearest the spot:
22,250. The SDK gives it to you.

Read down a column

Calls get cheaper as the strike rises; puts get dearer. Same shape as the simulator.

OI = contracts still open

Biggest here: 1.07 crore at the 22,400 call, 67.8 lakh at the 22,200 put.

Captured 9 Oct 2026, 00:21 IST on PROD with `MarketData.option_chain("NIFTY", exchange="NSE")`, after market close, so prices are 8 Oct's last trades. Nearest expiry, 13 Oct 2026 (the SDK's default). The SDK returns paise; shown in rupees. Indian digit grouping (lakh, crore).

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OI in one-liners

Open interest = contracts still open

Every contract has a buyer and a seller. OI counts the ones not yet closed or expired.

OI is not volume

Volume counts trades today. OI counts positions still alive at the end of the day.

Rising OI with rising price

New money is joining the move. Traders read it as **new longs** (not as proof of anything).

OI walls

A strike with very large OI (the 22,400 call on the last slide) is what **traders watch** as a "wall": a magnet or a barrier into expiry. A belief about behaviour, not a law of physics.

Phrase it the honest way: "**traders watch**", never "the market must".

Pass the underlying

```
chain = market_data.option_chain("NIFTY", exchange="NSE").chain

print(chain.current_price, chain.at_the_money_strike)
print(chain.all_expiries)
atm_ce = next(o for o in chain.ce
              if o.strike_price == chain.at_the_money_strike)
```

Per row

ref_id · strike_price ·
last_traded_price · iv · delta ·
gamma · theta · vega ·
open_interest · volume

Pass

"NIFTY", never an option symbol

Expiry

pick from `all_expiries`, do not
hardcode

The Greeks, as a car

Delta

Speed. Price change per 1-point move in the underlying.

Gamma

Acceleration. How fast delta changes.

Theta

Fuel burn. Value lost each day, even parked.

Vega

Road conditions. Reaction to expected volatility (IV).

The SDK hands you all four on every row of the chain, live and historical. You do not calculate them by hand.

Polling vs streaming



Polling

Ask again and again: "price now?"

Like refreshing a cricket score page. Slow, wasteful, hits rate limits.

Streaming (WebSocket)

One open connection. The server **pushes** every update.

Like ball-by-ball radio commentary. This is how a machine watches.

STREAM

index

orderbook

greeks

ohlcv

option

GIVES YOU

live price tops for indices and stocks

live market depth

live option Greeks

the current candle, updating

live option chain for one expiry

YOU

LOGIN

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PRICE

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OPTIONS

LIVE

PORTFOLIO

EXCHANGE

Register a **callback**

```
from nubra_python_sdk.ticker import websocketdata

def on_market_data(msg): print("[LIVE]", msg)

socket = websocketdata.NubraDataSocket(
    client=nubra, on_market_data=on_market_data)
socket.connect()
socket.subscribe(["NIFTY", "HDFCBANK"],
                 data_type="index", exchange="NSE")
socket.keep_running() # blocks this thread
```

Callback

Your function runs once per update. Keep it fast.

Subscribe

A list of keys plus a **data_type**; unsubscribe frees the budget.

keep_running

Blocks the thread. Use a thread if the program does other work.

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BOX 7 · EVERY PIPE HAS A LIMIT

Rate limits

WEBSOCKET STREAM	WEIGHT PER KEY
option (one expiry)	20
orderbook	5
ohlcv	2
index	1
greeks	1

50,000

weight per session. Over it, the subscription is rejected; unsubscribing frees weight.

Historical data

60 requests a minute, up to 5 symbols per request. Pace your loops.

Trading

10 operations a second on PROD · **100** on UAT (place, modify, cancel).

Per IP address

One college Wi-Fi = one public IP. 80 laptops share one budget. Stagger your calls, or stream instead of polling.

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The other half

Positions

Open trades today and their profit or loss

Holdings

Shares sitting in your demat account

Orders

Your instruction to the exchange. **UAT only** in these sessions.

Reading the market is half the machine. Acting on it is the other half. **Session 2.**

RECAP

Every box you just met



Next: Nubra's open source, live. Then, a quiz.

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Nubra **OSS**, live

nubra.io/products/api/apidashboard/nubra-oss

- ◆ **NubraOSS**: an open-source, customisable trading dashboard (beta). Source is public: github.com/socials-zanskar/NubraOSS
- ◆ Six modules: **No-Code** strategy builder · **Volume Breakout** scanner · **Webhook** strategies · **Scalper** · **Backtest Lab** · Risk & Tools (preview soon)
- ◆ Runs on your laptop: download the ZIP, run **Setup NubraOSS.cmd**, then **Run NubraOSS.cmd**, log in with OTP + MPIN

Also open

nubra-python-sdk: the SDK you installed. MIT licence, 105 runnable examples.
github.com/NubraAPI/nubra-python-sdk

What to try first

Search one stock, then open Backtest Lab and run one simple strategy (the page's **own first-run checklist**).

Verified 9 Oct 2026: nubra.io/products/api/apidashboard/nubra-oss (page text) and github.com/socials-zanskar/NubraOSS (public repo: FastAPI backend, React + Vite frontend; no licence file shown on the repo page). SDK repo: MIT, 105 examples.

ZERO TO PULL REQUEST

Four steps. No genius needed.

STAR

and fork the repo



RUN

one example locally



IMPROVE

a typo, a comment, a new example



PULL REQUEST

say what you changed

Easy first wins

Fix a typo · clarify a comment ·
add a missing example

Bring your Session 2

The project you build tomorrow
can become an example.

Where

github.com/NubraAPI/nubra-python-sdk ·
github.com/socials-zanskar/NubraOSS

BLOCK 5 · SHOWDOWN

DALAL STREET SHOWDOWN

8 questions. Fastest correct answer wins.

backboard-crafter-closure.ngrok-free.dev

Same login as the survey (USN + PIN). Keys 1 to 4 work too.

How it works: open the link · log in · 8 questions, 4 options each · 20 seconds a question · speed-scored: right and fast beats right and slow.

RECAP

Today in **four lines**

- ◆ Markets went from **shouting** to **screens** to **machines**
- ◆ A machine sees the market as **data**: price, depth, candles, options, live streams
- ◆ The Nubra SDK gives you all of it in **Python**
- ◆ Open source is how you **join in**

AFTER THE BREAK

Now **we build**

One project

All of us, together, from an empty file to a working market tool.

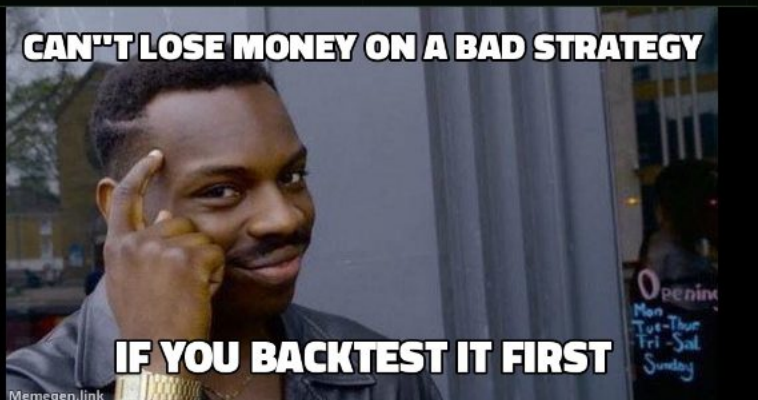
Your laptop

Python installed, credentials ready. Sandbox only, no real money.

You keep it

The code is yours to extend, and to contribute.

CAN'T LOSE MONEY ON A BAD STRATEGY



BEFORE THE BREAK

Is your **setup** ready?

- ◆ Python 3 installed `python --version`
- ◆ SDK installed `python -m pip install nubra-sdk`
- ◆ Nubra credentials in hand (phone number + MPIN)
- ◆ Laptop charged · charger packed

Stuck? Find a volunteer during the break. No Nubra account? Pair up with someone who has one, or use the offline data: nobody gets left out.

KEEP GOING

Docs • **code** • **contact**

Docs

nubra.io/products/api/docs

Code

github.com/NubraAPI/nubra-python-sdk

Say hi

DataVerse: [dataverse_bmsit](https://dataverse.bmsit.com)

Break time. See you in Session 2.